

6(a)	path/direction in which a (free) <u>positive</u> charge will move	B1
6(b)	(lines) closer together in Y/further apart in X	B1
6(c)(i)	$a = Eq / m$ or $F = Eq$ and $F = ma$	C1
	ratio = $(1e / 0.15 \text{ u}) \times (4 \text{ u} / 2e)$ or $1 / 0.15 \times 4 / 2$	C1
	ratio = 13	A1
6(c)(ii)	down quark charge is $-(1/3)e$	C1
	$-(1/3)e + q = -1e$ so $q = -(2/3)e$	A1
	$-(2/3)e$ is) anti-up / $\bar{u}$ (quark) (<i>allow charm or top antiquark</i>)	B1